

Experiment 2 Report

Contents:

Design a light control circuit with 74LS160, 74LS138 (3-8decoder) and necessary logic gates.

Requirement:

1. Red, green and yellow lights are required to change state in the order specified in the following table under the control of clock signal. 1 means "ON", 0 means "OFF".
2. Should be a self starting circuit.
3. All three lights need to connect to LED lights (Red, green or yellow)
4. 74LS160 should be connected to a seven-segmentation LED to show the calculation process.

CP顺序	红	黄	绿
0	0	0	0 ←
1	1	0	0
2	0	1	0
3	0	0	1
4	1	1	1
5	0	0	1
6	0	1	0
7	1	0	0
8	0	0	0 —

Figure 1 requirement table

Analysis:

1. Consider the output is the loop with 8 states, we can use 3 lower bit to design and implement an 8-ary counter.

We use asynchronous method to implement this function, we can just connect Q_3 with an inverter and then connect to $\overline{CR}$. Also, we need to connect the END, ENT, LORD with 1. The state transformation table is as follow:

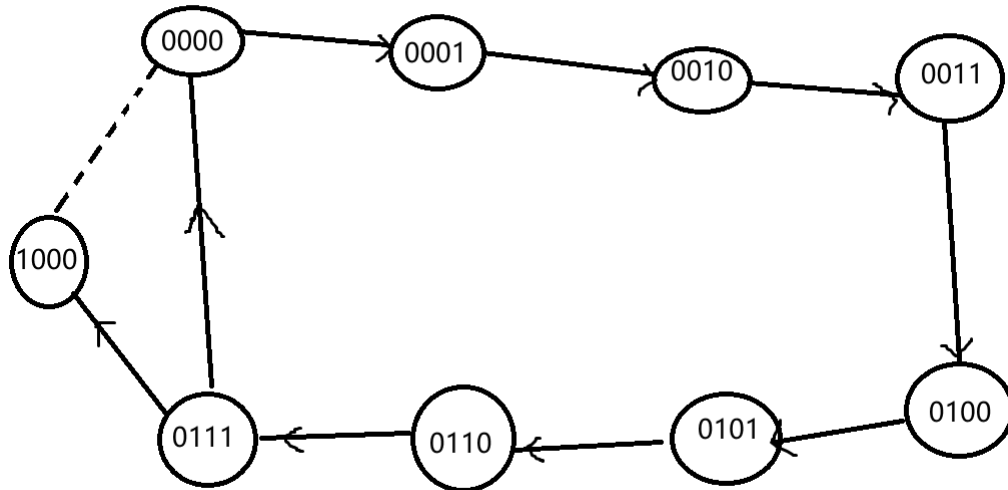


Figure 2 state transformation table

As the graph shown, this is a Self-starting, asynchronous. 8-ary counter, which satisfies our requirement.

- Here, we use R, Y, G represent “red”, “yellow”, and “green” respectively. Therefore, we can know the truth table of output state of counter Q_2, Q_1, Q_0 and R, Y, G, which is as shown below.

Q_2	Q_1	Q_0	R	Y	G
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	1	0	0

Table 1. The truth table

- According to the truth table, we can generate the logic expression:

$$R = \overline{Q_2} \overline{Q_1} Q_0 + Q_2 \overline{Q_1} \overline{Q_0} + Q_2 Q_1 Q_0$$

$$Y = \overline{Q_2} Q_1 \overline{Q_0} + Q_2 \overline{Q_1} \overline{Q_0} + Q_2 Q_1 \overline{Q_0}$$

$$G = \overline{Q_2} Q_1 Q_0 + Q_2 \overline{Q_1} \overline{Q_0} + Q_2 \overline{Q_1} Q_0$$

4. Considering given 74LS138, the 3-8 decoder, we can give the logic function of it:

$$F_0 = \overline{A} \overline{B} \overline{C}$$

$$F_1 = \overline{A} \overline{B} C$$

$$F_2 = \overline{A} B \overline{C}$$

$$F_3 = \overline{A} B C$$

$$F_4 = A \overline{B} \overline{C}$$

$$F_5 = A \overline{B} C$$

$$F_6 = A B \overline{C}$$

$$F_7 = A B C$$

Therefore, we have:

$$R = F_1 + F_4 + F_7$$

$$Y = F_2 + F_4 + F_6$$

$$G = F_3 + F_4 + F_5$$

Refer to this we can connect the lines, and implement the functions.

Implementation

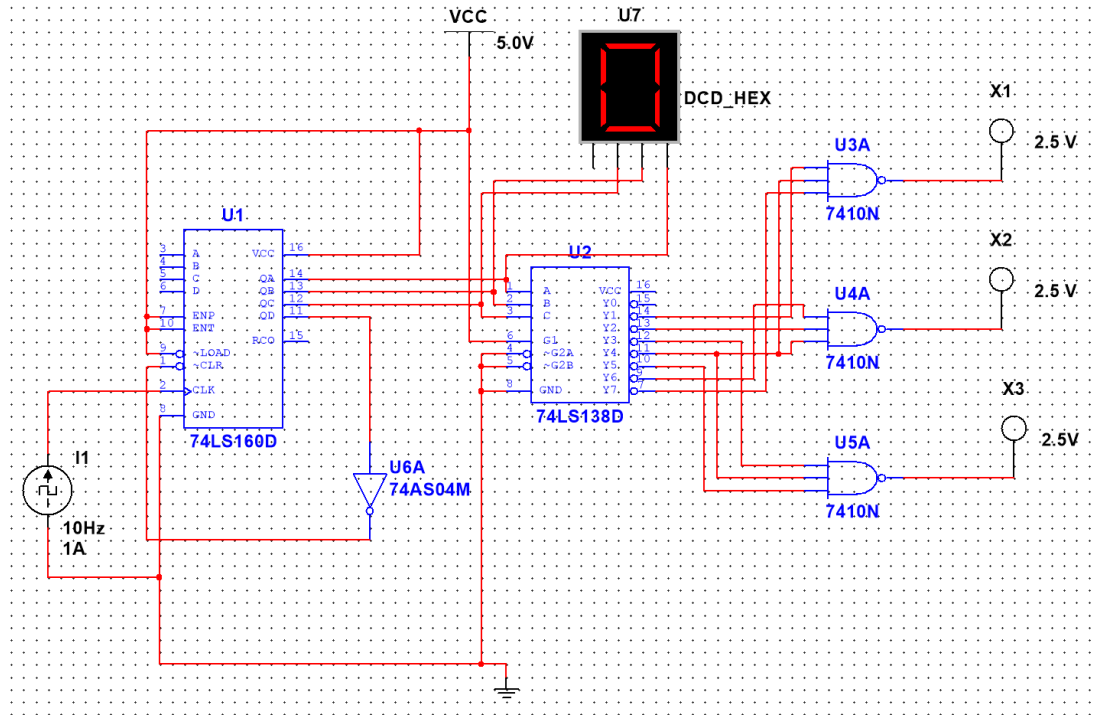
1. For the counter 74LS160, in addition to connecting the output pins QA, QB, and QC to the A, B, and C pins of the 74LS138 respectively, and connecting QD through a NOT gate to the CLR pin, the END, ENT, and LORD pins should be set to 1 (which is represented by connecting them to the power supply in the circuit). Additionally, the CLK pin should be connected to a clock current source, CLOCK_CURRENT, to ensure the proper operation of the counter, with the other end of the current source grounded.
2. Due to the characteristic of the 74LS138 outputting a low level, we convert the expression into a NAND-type logic with a low-level output.

$$R = \overline{\overline{F_1} \overline{F_4} \overline{F_7}}$$

$$Y = \overline{\overline{F_2} \overline{F_4} \overline{F_6}}$$

$$G = \overline{F_3} \overline{F_4} \overline{F_5}$$

And then we connect R,Y,G with the correspondent gate. We can get the circuit in Multisim as shown:



In the graph, x1 is red, x2 is yellow, and x3 is green.

Test &Result:

Run the circuit, we can see the 3 lights is on, when it is in specified condition shown as follows.

DCD_HEX	RED	YELLOW	GREEN
0			
1	ON		
2		ON	
3			ON
4	ON	ON	ON
5			ON
6		ON	
7	ON		

After that the DCD_HEX return to 0, and begins a new loop, which means our circuit is right to the requirements.

Comments:

In this experiment, I learned how to design the sequential logic circuit, and how to use some new chips, such as DCD_HEX, clock source, etc. This helps me understand sequential logic better. Hope I can get good grades in the final.